# IEEE 14-bus operation panel

Analysed hour

7 h

0 h 23 h

Restart step-through

**STATUS: Normal** All limits respected · 07:00 · 0 of 24 hours with violations

| Load | Generation | Losses (4.6%) | Min. load-bus V (B4) | Max. load-bus V (B9) | Max. loading (T1) |
|---|---|---|---|---|---|
| 243 MW | 254 MW | 11.7 MW | 1.019 pu | 1.059 pu | 48% |

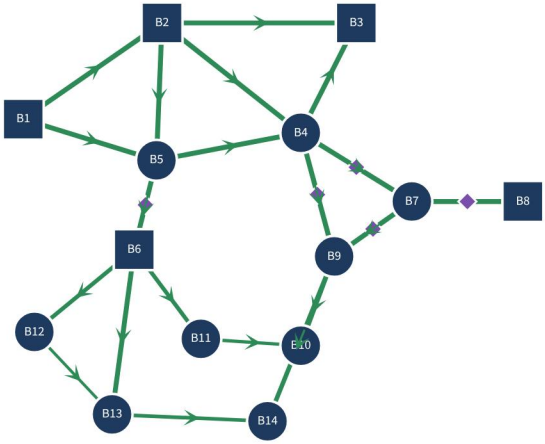
› 🔧 Laboratory: corrective actions

## 24-hour power flow

no violations in 24 h

Network | Load × generation | Branch loading | Voltages

Network state at 07:00.




Square = generator bus · circle = load bus · red = load-bus voltage outside limits · branch colour and width = loading (green, yellow, orange, red) · arrow = active-power direction · purple diamond = transformer.

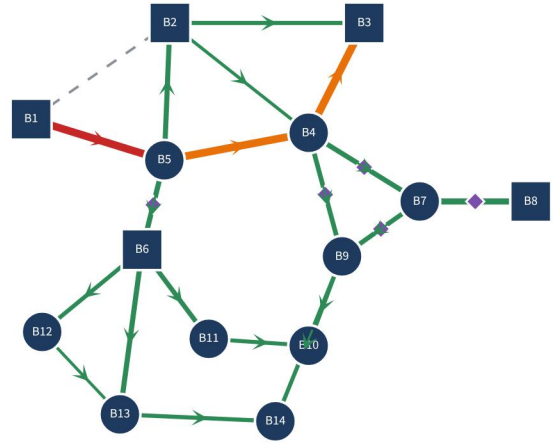
### Most loaded branches at this hour

## N-1 contingency

NOT secure · 1 violation(s)

Removed branch: L1 (1-2) · analysed hour: 07:00

Network | Violations | Before × after | Ranking




Square = generator bus · circle = load bus · red = load-bus voltage outside limits · branch colour and width = loading (green, yellow, orange, red) · arrow = active-power direction · purple diamond = transformer. Dashed branch = removed.

4 generator(s) at a reactive-power limit (PV bus switched to PQ).